

ADINet: An Adaptive Dependency Interaction Network for Arc Fault Detection

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Abstract—Deep learning has been successfully applied to arc fault detection for recognizing electrical abnormality and hence avoiding fires and equipment damage, but existing deep detection models have not fully considered the time-frequency feature interaction and suffer from high computational complexity. To address these issues, this paper proposes an Adaptive Dependency Interaction Network (ADINet) for arc fault detection. ADINet first generates the feature representation for a given input signal through three feature processing modules (feature extraction, feature fusion, and feature interaction) and then detects the signal as arc fault or not by the classification module, where feature fusion and interaction modules are newly designed. The feature fusion module uses lightweight Low-rank Fully Connected (LrFC) layers to fuse multi-band features extracted from original arc signals by the feature extraction module. The feature interaction module implements the information interaction in terms of time and frequency-band sequentially and adaptively incorporates the cross-band interaction information into the interacted time features. Experimental results on a public arc detection dataset demonstrate that ADINet consistently achieves the best detection performance across various resolutions and load configurations, offering an effective solution for arc fault detection in complex power usage scenarios.

Index Terms—Arc fault detection, Deep learning, Wavelet transform, Time-frequency information.

I. INTRODUCTION

ARC faults are a common and hazardous type of electrical abnormality, typically caused by wire aging, poor contact, or insulation degradation. They are characterized by high-frequency fluctuations and stochastic discharge components within the current signal. If these faults have not detected promptly, they can easily trigger electrical fires, damage equipment, and pose risks to personal safety, thereby threatening both distribution networks and household electrical security.

At present, arc fault detection methods can be divided into traditional and machine learning methods. Traditional arcing fault detection methods primarily rely on handcrafted feature extraction and signal-processing techniques. These approaches can be implemented by three ways: time-domain analysis [1]–[4], frequency-domain analysis [5]–[8], and time-frequency analysis [9]–[12]. Although these methods perform well under certain specific load conditions, they suffer from two main limitations. First, the performance of traditional approaches heavily depends on the quality of extracted features, which

are often tailored to a limited range of load types. When faced with emerging appliances or complex load combinations, these manually designed features typically exhibit poor generalization capability. Second, these methods are highly sensitive to noise, resulting in substantial degradation in detection accuracy under real-world conditions.

Machine learning methods have been widely applied to arc fault detection, including some statistical learning methods (such as, Support Vector Machines (SVM) [13]), and network learning models (such as, full connected neural network [14], [15]). Although statistical learning methods have been shown to improve detection performance to some extent compared to traditional methods, they still rely on manually designed features and exhibit limited performance when handling high-dimensional and non-stationary arc signals. As noted by Wang et al. [14], statistical learning methods generally exhibit inferior detection performance compared to deep learning approaches, especially under complex load conditions and data imbalance scenarios. Fortunately, network learning, especially for deep learning, has demonstrated significant advantages in arc fault detection. The deep learning model is mainly based on three kinds of basic network architectures: Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Transformer. Existing deep detection models use these architectures alone or in combination. Generally, deep models can automatically learn features from their inputs. Thus, according to the role of deep models, deep detection models can be grouped into two categories: feature extraction and feature re-extraction. The former extracts features directly from raw signals, while the latter first applies some specific data transformation, such as wavelet transform, and then re-extracts features using deep models.

As feature extraction methods, CNN, RNN and Transformer have been separately applied to arc fault detection. Wang et al. [16] designed an Arc Network (ArcNet) that is a CNN-based arc fault detection model. ArcNet learns the temporal information in an end-to-end way and achieves high accuracy even under high sampling rates through a one-dimensional (1D) CNN (1D-CNN) without any intermediate transformation. Zhou et al. [17] applied a customized Multi-Scale CNN (MS-CNN) to learning temporal features at different receptive fields from original arc signals. Wang et al. [18] proposed a Signal-Adaptive Feature Network (SAFNet) that uses a projection-pursuit mapping procedure to transform 1D current signals into two-dimensional (2D) matrices and incorporates depth-wise separable convolution together with a bottleneck structure to achieve lightweight detection. Li et al. [19] directly adopted RNN to capture temporal dependencies from original signals and combined it with a rapid continuous detection

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strategy and probability correction mechanism, performing exceptionally well in processing dynamic arc signals. Chabert et al. [20] first introduced Transformer Neural Network (TNN) to detect arc faults, providing the earliest evidence that the Transformer architecture is feasible and effective for this task. Although the above models are simple and effective, they ignore the time-frequency nature of arc signals and then may fail to capture the relationship across-band in signals.

To fully use the frequency or time-frequency information contained in signals, researchers have considered the feature re-extraction method. Such re-extraction methods first apply some data pre-processing techniques, such as low-pass filter and wavelet analysis, to original signals and then predict arc faults through deep models on the processed signals. For example, a Transfer Learning-based Low-Energy arc-fault Detection network (TL-LED^{arc}Net) takes signals filtered by a 20kHz low-pass filter as the inputs of 1D-CNN [21]. Yang et al. [22] proposed a Temporal Domain Visualization Convolutional Neural Network (TDV-CNN) that first handles gray images generated from raw signals through a 5–200 kHz Butterworth digital band-pass filter and then detect arc faults based on filtered images by a CNN. Similarly, Chu et al. [23] designed a High-Frequency Coupling Convolutional Neural Network (HCCNN), which employs a high-frequency coupling sensor to convert the arc signals into gray images, strengthening the representation of arc-related features. The CNN-Transformer Parallel Neural Network (CTPNN) proposed by Ning et al. [24] first obtains low- and high-frequency components through ADC (Analog-to-Digital Converter) sampling on the measured data, and then uses CNN and Transformer branches to extract features from low- and high-frequency signals, respectively. Both Yu et al. [25] and Zhang et al. [26] first pre-processed raw signals using wavelet decomposition and then generated the data containing time-frequency information. Further, the former adopted GRU (Gated Recurrent Unit), a variant of RNN, to capture multi-resolution time-frequency features and temporal evolution patterns, while the latter employed Residual Network (ResNet, a kind of CNN) to maintain robust detection performance under varying and dynamic load conditions. Overall, feature re-extraction methods emphasize frequency or time-frequency information contained in raw signals, but most methods pay little attention to deep models themselves. Thus, these feature re-extraction methods inherit the high computational cost due to the complexity of deep models. Despite the time-frequency nature of arc signals has been considered, none of these studies has concerned about the time-frequency information interaction, which restricts their representational ability for arc fault detection.

To interact time-frequency features and alleviate the high computational complexity, we propose an Adaptive Dependency Interaction Network (ADINet) for arc fault detection. The main contributions of this work are summarized as follows:

- 1) A Low-rank Feature Fusion (LFF) module is proposed to fuse multi-band features obtained from arc signals. This module employs lightweight Low-rank Fully Connected (LrFC) layers to achieve efficient feature reconstruction

and dimensional alignment, thereby preserving feature representation capability.

- 2) A Time-Frequency Feature Interaction (TFFI) module is presented for interacting the time-frequency feature information and then generating the interacted time-frequency representation. In this module, a newly designed Adaptive Dependency Interaction (ADI) block is used to interact time and frequency-band information in turn. In addition, ADI can adaptively adjust the strength of cross-band interaction information by a Band Adaptive Unit (BAU) and the proportion of the cross-band interaction information injected into the interacted time features through a Band Gate Unit (BGU).
- 3) Based on LFF and TFFI modules, we propose ADINet for arc fault detection. The idea behind ADINet is to classify the given arc signals into one of two types (normal and fault) by handling time-frequency information contained in these signals. Comprehensive comparison with mainstream models was conducted on a public arc fault dataset. Results show that ADINet has a small number of parameters and the superior detection performance in both accuracy and F1-score. In short, ADINet is a lightweight and efficient detection model.

The rest of this paper is organized as follows. Section II introduces the proposed ADINet in details, including its architecture and main modules. Section III provides experimental results on arc fault detection, while Section IV makes a summary of this paper and gives a guidance of future work.

II. ADAPTIVE DEPENDENCY INTERACTION NETWORK

A. Architecture of ADINet

Fig. 1 shows the overall architecture of ADINet. This model has four main modules: feature extraction (1D discrete wavelet transform, 1D-DWT), feature fusion (LFF), feature interaction (TFFI), and classification (Softmax). Given an input signal $\mathbf{x}$, the goal of ADINet is to automatically identify whether it is normal or arc fault. In the following, we simply describe the inference process.

First, the feature extraction module is to capture both steady-state and transient components from the current signal $\mathbf{x}$ by applying 1D-DWT. After this module, we have multi-band feature signals, including the low-frequency feature vector $\mathbf{a}_J$ and multi-scale high-frequency vectors $\mathbf{d}_J, \dots, \mathbf{d}_1$. The LFF module uses Low-rank Fully Connected (LrFC) layers to fuse these multi-band feature signals and then produce a spatially consistent fused representation. After that, the TFFI module is to implement the time-frequency information interaction based on the fused representation, which can process patches generated by the fused representation in parallel and thus result a small computational cost. Finally, the Softmax classifier takes the interacted representation obtained by the TFFI module as the input and then outputs the category prediction of $\mathbf{x}$.

B. DWT Feature Extraction

To fully dig up the multi-scale time-frequency information contained in current signals, the feature extraction module

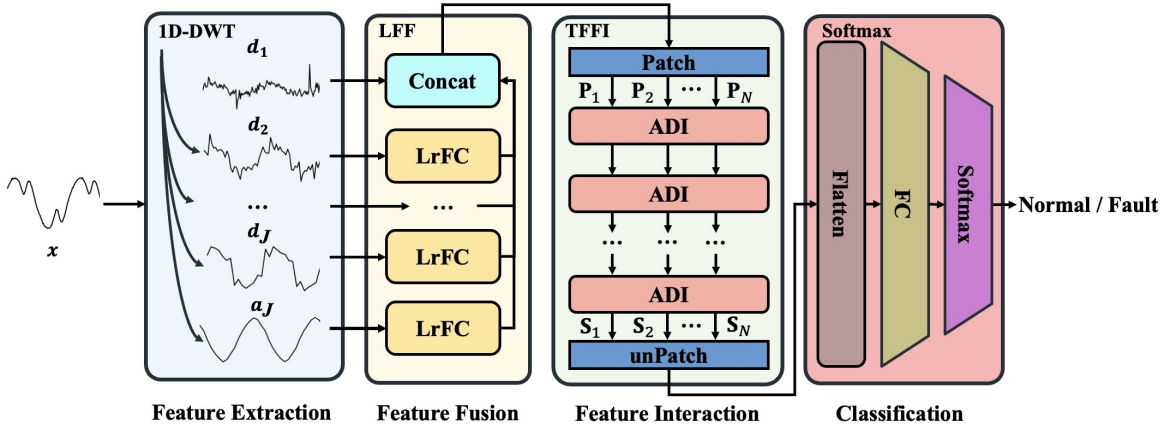


Fig. 1: Overall architecture of the proposed ADINet, consisting of four main modules: (1) 1D-DWT feature extraction, which performs multi-band decomposition using 1D-DWT; (2) LFF, which is to fuse multi-band feature vectors; (3) TFFI, which is responsible for interacting the time-frequency information; (4) Softmax classification, which judges whether an input is normal or fault.

utilizes 1D-DWT. This choice is further validated by the ablation study in Section III-D. Let $\mathbf{x} \in \mathbb{R}^L$ be the current signal sequence. Through 1D-DWT, we can decompose $\mathbf{x}$ into approximation coefficients $\mathbf{a}$ and detail coefficients $\mathbf{d}$, where $\mathbf{a}$ represents the energy distribution and steady-state load characteristics of $\mathbf{x}$ in the low-frequency range, and $\mathbf{d}$ corresponds to transient fluctuations and pulse components of $\mathbf{x}$ caused by arc discharges in the high-frequency band. Without loss of generality, let J be the number of decomposition levels. Then, we can have the set $\mathcal{M}$ of multi-band features. Namely,

$$\mathbf{x} \xrightarrow{\text{1D-DWT}} \mathcal{M} = \{\mathbf{a}_J, \mathbf{d}_J, \mathbf{d}_{J-1}, \dots, \mathbf{d}_1\} \quad (1)$$

where $\mathbf{a}_J$ denotes the low-frequency approximation coefficients at level J , and $\mathbf{d}_j$ denotes the high-frequency detail coefficients at level j . Note that there is one low-frequency and J high-frequency feature vectors in $\mathcal{M}$.

Fig. 2 illustrates original signals and their multi-band features obtained by a three-level Haar wavelet transform, or $J = 3$. In Figure 2a, the normal current exhibits a stable and regular energy structure across all scenes. Conversely, Figure 2b implies that the arc signal displays pronounced non-stationary characteristics with sharp spikes and pulse peaks, particularly evident in high-frequency detail components. According to Fig. 2, it is sufficient to distinguish the high-frequency disturbances and low-frequency trends in arc signals through a three-level decomposition. The value of J is further discussed in the hyper-parameter analysis of Section III-B.

C. Low-rank Feature Fusion

After the feature extraction module, we have the set of multi-band features, or $\mathcal{M} = \{\mathbf{a}_J, \mathbf{d}_J, \dots, \mathbf{d}_1\}$, where $\mathbf{a}_J \in \mathbb{R}^{L/2^J}$, $\mathbf{d}_j \in \mathbb{R}^{L/2^j}$, $j = 1, \dots, J$. The goal of LFF module is to fuse these multi-band features and form a spatially consistent representation. As shown in Figure 1, this LFF module contains J LrFC blocks and a concatenation (Concat) operation.

Note that the lengths of multi-band feature vectors are not equal to each other. Thus, we need to align their lengths first.

To preserve the greatest extent of original signal information, we set the length of $\mathbf{d}_1$ as the reference one, which is the longest feature vector among vectors in the set $\mathcal{M}$. Through J LrFC blocks, we separately map the other feature vectors into the $L/2$ -dimensional space. Specifically, a LrFC block consists of two fully connected (FC) layers with an intermediate nonlinear activation, which can be expressed by

$$\text{LrFC}(\mathbf{m}) = \text{FC}(\text{FC}_\phi(\mathbf{m})) \quad (2)$$

where $\mathbf{m} \in \mathbb{R}^\ell$ is an arbitrary vector in $\mathcal{M}$, ℓ is the length of $\mathbf{m}$, $\text{FC}(\cdot)$ is the FC layer without activation function, $\text{FC}_\phi(\cdot)$ is the FC layer with the GELU (Gaussian Error Linear Unit) activation function having the form

$$\text{GELU}(z) = \frac{z}{2} \left(1 + \frac{2}{\sqrt{\pi}} \int_0^{\frac{z}{\sqrt{2}}} e^{-t^2} dt \right) \quad (3)$$

In each LrFC block, FC_ϕ maps the input vector from $\mathbb{R}^\ell$ to $\mathbb{R}^{\ell/2}$, while the subsequent FC layer projects it to the target space $\mathbb{R}^{L/2}$. Therefore, the LrFC block performs a learnable linear projection from a lower-dimensional space to a higher-dimensional space, enabling flexible alignment of feature lengths across different frequency bands.

Note that $\ell < L/2$ in our scenario. Thus, the main role of LrFC is not only to preserve the core information of the frequency-band features but also to increase the length of the input vector with a low computational cost. In this way, both the steady-state load information and the transient arc disturbance information from multi-bands can be effectively reconstructed in a unified feature space.

Assume that the aligned feature vectors of $\mathbf{a}_J$ and $\mathbf{d}_j$ are $\hat{\mathbf{a}}_J$ and $\hat{\mathbf{d}}_j$, respectively, $j = 2, \dots, J$. Subsequently, these aligned feature vectors and $\mathbf{d}_1$ are concatenated along the channel (or frequency-band) dimension, which forms a fused feature representation. That is,

$$\mathbf{U} = \text{Concat}(\hat{\mathbf{a}}_J, \hat{\mathbf{d}}_J, \dots, \hat{\mathbf{d}}_2, \mathbf{d}_1) \in \mathbb{R}^{C \times L/2} \quad (4)$$

where $\text{Concat}(\cdot)$ is the concatenation operation, and C denotes the number of frequency bands. In this case, $C = J + 1$.

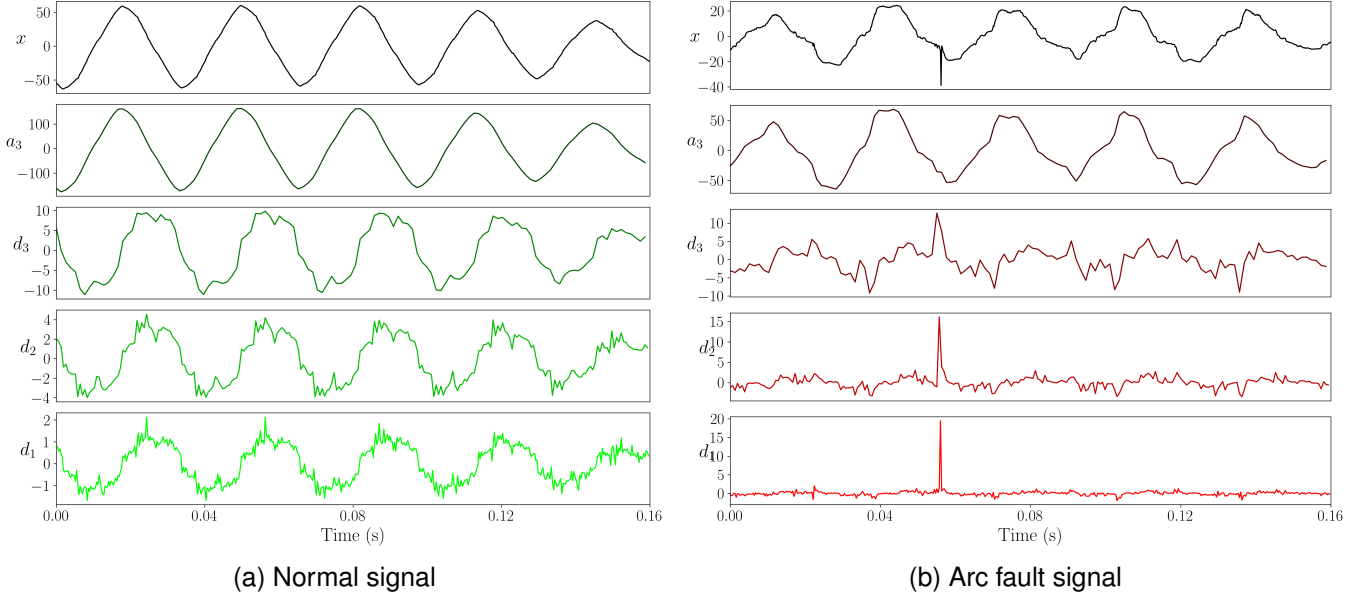


Fig. 2: Time-frequency decomposition of normal (a) and arc fault (b) current signals using a three-level Haar wavelet. On the left side, the approximation feature vector $\mathbf{a}_3$ exhibits a smooth and periodic waveform corresponding to the stable load operation, while the detail features $\mathbf{d}_3$, $\mathbf{d}_2$, and $\mathbf{d}_1$ remain relatively regular with slight fluctuations. On the right side, although $\mathbf{a}_3$ still preserves the fundamental current trend, noticeable waveform distortion and local irregularities emerge. In contrast, the detail components $\mathbf{d}_3$ and $\mathbf{d}_2$ present pronounced non-stationary fluctuations, and $\mathbf{d}_1$ contains sharp impulsive peaks, reflecting intense and high-frequency disturbances induced by unstable arc discharge.

This fused feature representation $\mathbf{U}$ preserves both the global trends and steady-state load information reflected by the low-frequency components, as well as the transient arc disturbances captured by the high-frequency components.

D. Time-Frequency Feature Interaction

In the situation of time-frequency analysis induced by DWT, we notice potential information interactions across different frequency-bands decomposed from the given current signal. Intuitively, for example, the low-frequency feature vector $\mathbf{a}_3$ on the third level is complementary to the high-frequency feature vector $\mathbf{d}_3$ on the same level because $\mathbf{a}_3$ and $\mathbf{d}_3$ can be jointly reconstructed to the low-frequency feature vector $\mathbf{a}_2$ on the second level. Actually, all frequency-band features are independent parts of representations for the input. Therefore, we need a richer representation that can reflect the time and frequency interactive information of current signals, which is the goal designing the TFFI module.

The design of TFFI is inspired by the Dual Dependency Interaction (DDI) block that was proposed by Hu et al. [27]. DDI can make an interaction along temporal and channel dimensions across different scales. However, the fixed scaling factor of DDI cannot dynamically and adaptively adjust the strength and injected proportion of interacted channel information for different inputs. The redesigned TFFI module not only inherits the data procedure of DDI but also solves the issue of DDI.

As shown in Figure 1, the TFFI module has mainly three kinds of blocks: patch, ADI, and unpatch. First, this module

divides the fused feature representation $\mathbf{U}$ into N patches, and then, all original patches are fed into the ADI blocks. Finally, we reassemble these interacted patches into the time-frequency representation with the interaction information along time and band dimensions.

1) *Patch*: Generally, there exists local correlation in current signals. In other words, the signal value at the k -th time would be related to values at previous times, and has an effect on values at subsequent moments. Thus, the local time interaction is more important than the global time interaction in our TFFI.

The patch block takes the fused representation $\mathbf{U}$ as its input and outputs multiple patches. That is,

$$\mathcal{P} = \text{Patch}(\mathbf{U}) \quad (5)$$

where $\mathcal{P} = \{\mathbf{P}_1, \dots, \mathbf{P}_N\}$ is the set of patches, $\mathbf{P}_k \in \mathbb{R}^{C \times P}$, P is the length of time window (also called the patch length), N is the number of patches, and $L = 2NP$. Note that the row vector of the patch $\mathbf{P}$ is a time segment with respect to a frequency-band, and the column vector of $\mathbf{P}$ is a cross-band feature vector with respect to a certain moment.

2) *ADI*: After the patch block, we input all patches $\mathbf{P}_k$ ($k = 1, \dots, N$) into the first ADI block in parallel. Note that there are n ADI blocks in series. The ADI block is the key component in TFFI, which implements the information interaction along time and frequency dimensions, and enables the model to dynamically adjust both the strength and injected proportion of cross-band interaction information. Fig. 3 shows the structure of one ADI block, which consists of two phases: time and frequency interaction.

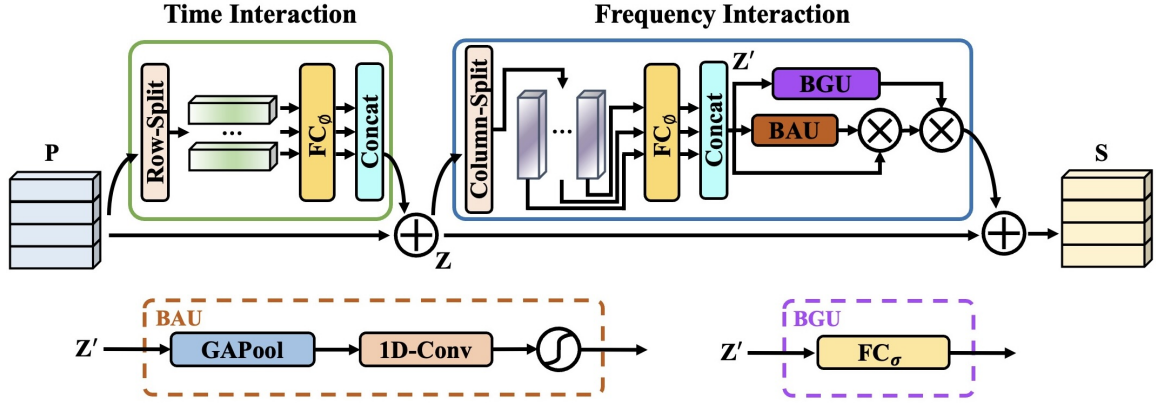


Fig. 3: Structure of an ADI block.

First, the time interaction phase is to interchange the time information contained in time segments, which is performed independently on each band. In doing so, we can extract the local time information within each band. For any input $\mathbf{P}$ of the ADI block, we split it in row and then have C feature vectors, or $\mathbf{p}_i \in \mathbb{R}^P$, $i = 1, \dots, C$. These vectors are fed into an FC layer in parallel, which implements the information interaction intra-segments. The interacted time feature representation $\mathbf{Z}$ can be expressed as

$$\mathbf{Z} = \mathbf{P} + \text{Concat}(\text{FC}_\phi(\mathbf{p}_1), \dots, \text{FC}_\phi(\mathbf{p}_C)) \in \mathbb{R}^{C \times P} \quad (6)$$

where $\text{FC}_\phi(\cdot)$ has P neurons, which can make the lengths of input and output the same.

Second, the frequency interaction phase is to fuse the frequency information across frequency-bands without considering time segments. In doing so, we can extract the across-band information at a certain moment. The representation $\mathbf{Z}$ is the input of this phase. We split it in column and then obtain P feature vectors, $\mathbf{z}_j \in \mathbb{R}^C$, $j = 1, \dots, P$. These vectors are also handled by an FC layer to fuse the information across band. Then, the cross-band interaction information matrix can be described as

$$\mathbf{Z}' = \text{Concat}(\text{FC}_\phi(\mathbf{z}_1), \dots, \text{FC}_\phi(\mathbf{z}_P)) \in \mathbb{R}^{C \times P} \quad (7)$$

where $\text{FC}_\phi(\dots)$ contains C neurons. To achieve adaptive modulation and stable information fusion across frequency bands, we introduce BAU and BGU based on DDI, where BAU is responsible for adaptively adjusting the strength of cross-band interaction information, and BGU for calculating the proportion of cross-band interaction information infused into the time interacted representation. Both BAU and BGU are also shown in Figure 3.

In BAU, a global average pooling (GAPool) is first performed along the time dimension to extract statistical information for each frequency-band, and a 1D convolution (1D-Conv) is then used to implement a linear mapping. This process generates the adaptive band weight vector $\mathbf{w}_b$ that can be expressed as

$$\mathbf{w}_b = \text{BAU}(\mathbf{Z}') = \sigma(1\text{D-Conv}(\text{GAPool}(\mathbf{Z}'))) \in \mathbb{R}^C \quad (8)$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function, or

$$\sigma(t) = \frac{1}{1 + e^{-t}} \quad (9)$$

which constrains weights within the range $[0, 1]$. BAU is a variant of scaling factor in DDI, which allows the TFFI module to adaptively enhance the band information coupling if this band carries more information, ensuring a dynamic balance of the model under varying current inputs.

In BGU, we first make a linear transformation through the linear layer and the Sigmoid activation function, or FC_σ . Then, we have the gated weight vector $\mathbf{w}_g$, that is,

$$\mathbf{w}_g = \text{BGU}(\mathbf{Z}') = \text{FC}_\sigma(\mathbf{Z}') \in \mathbb{R}^C \quad (10)$$

BGU controls the proportion of coupled information injected along the frequency-domain dimension, and thus regulates the balance of information between the main path and the branch.

Finally, the ADI block outputs

$$\mathbf{S} = \mathbf{Z} + \mathbf{w}_g \odot (\mathbf{w}_b \odot \mathbf{Z}') \in \mathbb{R}^{C \times P} \quad (11)$$

where $\odot$ denotes the multiplication operation between vector and matrix, or the i -th element of a vector multiplies the i -th row elements of a matrix. With ADI blocks, we can dynamically enhance the response to key frequency-band features while maintaining the stability of time interaction information, thereby achieving more discriminative and physically meaningful time-frequency features.

3) *Unpatch*: After the ADI blocks, N patches can be transformed to N interacted patches $\mathbf{S}_i$, $i = 1, \dots, N$. Then, we integrate them into the interacted time-frequency representation $\mathbf{Y}$ via an unpatch block. The unpatch block rearranges and concatenates the features along the time dimension, and can be described as

$$\mathbf{Y} = \text{Unpatch}(\mathbf{S}_1, \dots, \mathbf{S}_N) \in \mathbb{R}^{C \times L/2} \quad (12)$$

This unpatch block preserves the local time interaction information within each patch while maintaining the global time structure of the multi-band signals.

E. Softmax classification and training loss

The classification module is designed to make a final decision on the label of current signal $\mathbf{x}$ based on the interacted representation $\mathbf{Y}$ generated by the TFFI module. Specifically, the interacted representation $\mathbf{Y}$ is first flattened along the time dimension to form a compact feature vector. This vector is then fed into two FC layers. Finally, a Softmax function is used to produce the probability distribution over the two classes corresponding to normal and arc fault conditions. Formally, the classification process can be expressed as

$$\hat{\mathbf{y}} = \text{Softmax}(\text{FC}(\text{FC}_\phi(\text{Flatten}(\mathbf{Y})))) \in \mathbb{R}^2 \quad (13)$$

where $\hat{\mathbf{y}} = [\hat{y}_1, \hat{y}_2]^T$ is the predicted class probability vector, $\hat{y}_1$ and $\hat{y}_2$ represent probabilities of the normal and fault classes, respectively, $\text{Flatten}(\cdot)$ is the flatten operation, and the function $\text{Softmax}(\cdot)$ has the form

$$\text{Softmax}(t_j) = \frac{e^{t_j}}{\sum_i e^{t_i}} \quad (14)$$

To train the proposed ADINet, the cross-entropy loss is employed as the optimization objective. This loss function evaluates the discrepancy between the predicted class scores and the ground-truth labels, and is well suited for classification tasks involving arc and non-arc current signals. Given an input pair $(\mathbf{x}, \mathbf{y})$ with the current signal $\mathbf{x}$ and its label vector $\mathbf{y} = [y_1, y_2]^T \in \mathbb{R}^2$, the loss function is defined as

$$\mathcal{L}_{\text{CE}}(\mathbf{x}) = -y_1 \log \hat{y}_1 - (1 - y_2) \log(1 - \hat{y}_2), \quad (15)$$

The classification module is intentionally kept simple to avoid introducing unnecessary computational overhead and to ensure that the performance gains primarily stem from the proposed time–frequency feature extraction, fusion, and interaction mechanisms.

III. EXPERIMENTS

A. Dataset Description and Model Evaluation

To validate the effectiveness of the proposed ADINet model in arc fault detection, experiments were conducted on the publicly available IAED (Intever public database for Arcing Event Detection) dataset. The dataset was proposed by Zhou et al. [17] and constructed by Southeast University and Jiangsu Intever Energy Technology¹. IAED contains arc current data collected from ten common household appliances under three typical load connection types, including single load, series–parallel dual loads, and parallel dual loads.

According to sampling rates, IAED is divided into High-Resolution (HR) and Low-Resolution (LR) sets. The HR data were segmented into sequences of 1000 points (approximately 100 ms), while the LR data were segmented into sequences of 1024 points (approximately 160 ms). Based on load connection types, HR and LR sets are further divided into four subsets: HR-single, HR-mix1, LR-single, and LR-mix2, as shown in Table I. The HR-mix1 and LR-mix2 subsets represent mixed-load conditions, and the HR-single and LR-single subsets

TABLE I: Description of the IAED dataset.

Subset	Sampling rate (kHz)	# Attribute	# Sample
HR-mix1	10	1000	22,430
HR-single	10	1000	11,526
HR-all	10	1000	33,956
LR-mix2	6.4	1024	6288
LR-single	6.4	1024	14,084
LR-all	6.4	1024	20,372

correspond to single-load scenarios. Note that HR-all and LR-all are HR and LR subsets themselves, respectively, which are used to assess overall generalization performance.

Following the data division strategy in [17], each subset was randomly split into training and test sets with a ratio of 80%/20%. The random partition was repeated five times, and the final results are reported as the mean and standard deviation over the five runs. Because arc fault detection is actually a binary classification task, we used two metrics F1-score and accuracy to evaluate the model performance.

B. Experimental Setup and Hyper-parameter Analysis

All experiments were conducted under consistent conditions. The model training and test in this study were performed under the PyTorch framework on an NVIDIA GeForce RTX 4060Ti GPU with 8 GB of memory. During training, GPU acceleration was utilized. General hyper-parameters were set as follows: the number of epochs is 300, the batch size is 2048, the initial learning rate is 10^{-3} , and the optimizer used is AdamW.

Customized hyper-parameters of ADINet are discussed as follows. The DWT feature extraction module has a special hyper-parameter J , the number of decomposition levels, which controls the number of frequency bands, or the amount of high-frequency detail information. In the TFFI module of ADINet, there are two customized hyper-parameters that need to be carefully set in advance: P (the length of time window, or patch length) and n (the number of ADI blocks). The length P reflects the scope of time information interaction and is related to the length of input signals, and n represents the depth of information interaction.

1) *Number of Decomposition Levels*: The impact of the hyper-parameter J on the model performance and complexity was evaluated on the HR-mix1 subset, where $J \in \{2, 3, 4, 5\}$, $P = 20$, and $n = 3$. Fig. 4 illustrates the relationship of both model performance (F1-score and accuracy) vs. decomposition level J on the left-hand vertical coordinate, and that of model complexity (parameter count) vs. J on the right-hand side.

From Fig. 4, we can see that both F1-score and accuracy always improve as the increase of decomposition level from 2 to 5. Additionally, the curve of model complexity vs. J has a trough at $J = 3$, or the smallest parameter count 0.86M.

Although increasing the decomposition level to $J = 4$ and $J = 5$ leads to slight improvements in the model performance, the parameter count increases significantly. Conversely, a

¹<https://dx.doi.org/10.21227/j9eg-mz75>

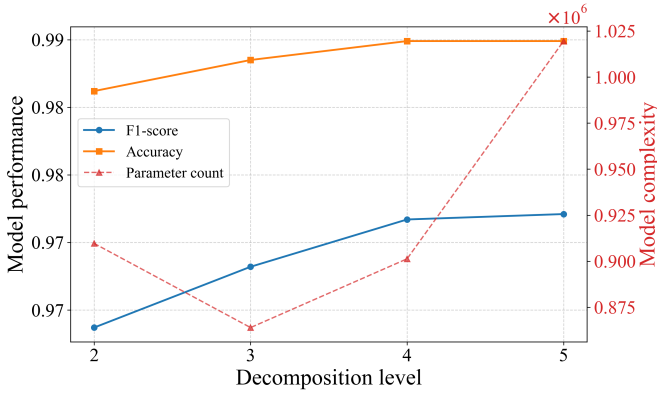


Fig. 4: Curves of model performance and complexity vs. decomposition level for the HR-mix1 subset, where the vertical coordinate on the left-hand side represents both F1-score and accuracy, and that on the right-hand side is the parameter count.

shallower decomposition at $J = 2$ results in higher parameter count (0.91M) and lower F1-score (96.87%), indicating insufficient separation of transient arc components from low-frequency load dynamics. Therefore, $J = 3$ is adopted as the optimal setting, achieving an effective balance between model compactness and detection capability.

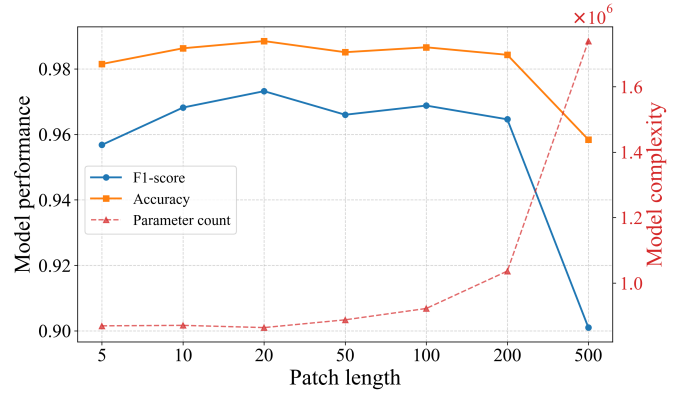
2) *Number of ADI Blocks*: To evaluate the impact of information interactive depth on the model performance, we conducted experiments on the number of ADI blocks using the HR-mix1 dataset, where n varies from 1 to 5.

The results are shown in Table II, where the patch length $P = 20$ and the decomposition level number $J = 3$. As the number of ADI layers increases from $n = 1$ to $n = 2$, the model performance gradually improves. A significant improvement is observed at $n = 3$, where the model achieves the best overall performance with an F1-score of 98.10% and accuracy of 99.15%. This indicates that a three-layer ADI structure achieves an optimal balance between capturing time-frequency interaction information and maintaining feature stability. Further increasing n to 4 or 5 results in the performance degradation, particularly in F1-score, as the excessively deep interaction introduces the redundant information and amplifies noise.

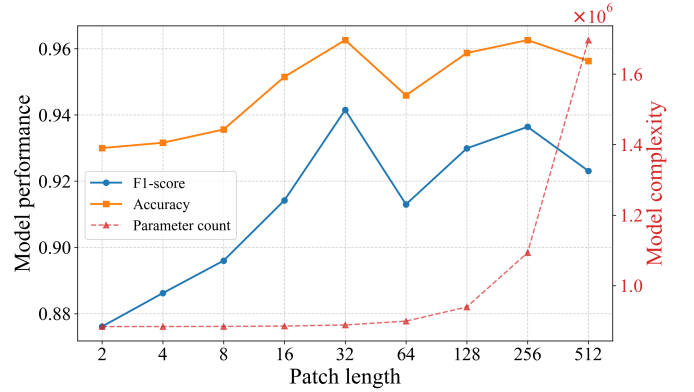
TABLE II: Hyper-parameter analysis on the number of ADI blocks on HR-mix1.

n	F1-score (%)	Accuracy (%)
1	97.72 ± 0.22	99.01 ± 0.09
2	97.79 ± 0.31	99.03 ± 0.13
3	98.10 ± 0.14	99.15 ± 0.06
4	97.67 ± 0.15	98.97 ± 0.06
5	95.54 ± 0.43	98.06 ± 0.16

Overall, the ADINet model with three ADI blocks achieves the best performance on the HR-mix1 subset. Thus, $n = 3$ represents the best depth setting and was used for following experiments.



(a) HR-mix1



(b) LR-mix2

Fig. 5: Curves of model performance and complexity vs. patch length P for the HR-mix1 subset (a) and the LR-mix2 subset (b), where each sub-figure has double vertical coordinates: the left-hand side representing both F1-score and accuracy and the right-hand side denoting parameter count.

3) *Patch Length*: To find a suitable value for the hyper-parameter P , we only varied it within a given range and investigated its effect on the model performance and complexity. Because the setting of P is related to the original feature length of inputs, we separately conducted experiments on both HR-mix1 and LR-mix2 that have different original feature numbers. The hyper-parameter P took a value in the range of $\{5, 10, 20, 50, 100, 200, 500\}$ and $\{2, 4, 8, 16, 32, 64, 128, 256, 512\}$ for the HR-mix1 and LR-mix2 datasets, respectively.

Fig. 5 plots curves of the model performance and complexity vs. P on two subsets. From Fig. 5a, we can see that the curves of model performance first slowly rise as the increase of P from 5 to 20 and then gradually decrease as the increase of P from 20 to 500. It is obvious that these curves all have a peak at $P = 20$ on the HR-mix1 subset. In this case, the model has the lowest parameter count 0.86M among all settings. Additionally, the number of parameters is proportional to the patch length P . The similar conclusion can be obtained by observing Fig. 5b. Undoubtedly, $P = 32$ yields the best model performance with remaining a relatively low parameter count

0.88M.

C. Comparison Results

To validate the effectiveness of the ADINet model in arc fault detection, we made a comparison with statistical learning methods (including SVM, K Nearest Neighbor (KNN), and Random Forest (RF)) that have been used for arc fault detection by Zhou et al. [17], and deep models (including ArcNet [17], GRU [25], TNN [20], and MS-CNN [17]) across multiple subsets of the IAED dataset.

Experimental results on the IAED dataset are summarized in Table III, where the best performance on one subset is highlighted in bold among all compared methods. On the HR subsets, statistical learning methods (SVM, KNN, and RF) are inferior to that of deep models in the whole, which indicates that statistical models have limited capability in representing complex arc-induced transient characteristics using fixed or shallow features, while deep models show clear advantages by extracting deep features. Notably, our ADINet consistently outperforms all baseline models across all three HR subsets, achieving the highest F1-score and Accuracy, which demonstrates its superior capability in representing original signals under high-resolution conditions.

Compared to the HR subsets, the performance of different methods becomes worse on the LR subsets. This decline underscores the models' sensitivity to lower sampling resolutions and higher load complexity. Under such conditions, the subtle time variations induced by arcing are smoothed and become challenging to distinguish from the background load dynamics. Among all compared models, ADINet maintains the most stable and superior performance on LR subsets, achieving the highest F1-score and accuracy. These results indicate that ADINet generalizes well across different load configurations and signal quality conditions because it can effectively mitigate the loss of time information caused by low sampling resolution.

D. Ablation Study

ADINet consists of four modules, where feature fusion (LFF) and feature interaction (TFFI) are newly designed. To evaluate these new modules in ADINet, we conducted ablation experiments on the HR-mix1 subset. First, we considered the cooperation of both modules, and then, investigated the internal structure of TFFI.

1) Comparison of Different Feature Extraction Methods:

To justify the selection of the Discrete Wavelet Transform (DWT) in the feature extraction stage, we conduct a comparative study with several commonly used signal transformation methods, including DFT (implemented via FFT), Power Spectral Density (PSD), Short-Time Fourier Transform (STFT), Continuous Wavelet Transform (CWT), Hilbert–Huang Transform (HHT), and Wavelet Packet Transform (WPT). The experiments are conducted on the HR-mix1 subset, and the results are summarized in Table IV.

As shown in Table IV, the proposed DWT-based method achieves the best performance, with an F1-score of 97.32%

and an accuracy of 98.85%, outperforming all other transformation methods. Among the compared approaches, DFT and STFT obtain relatively competitive results, indicating that these methods can effectively capture the time-frequency characteristics of arc signals. Methods such as CWT and WPT, although capable of providing multi-scale representations, show inferior performance compared to DWT. HHT exhibits the worst performance among all methods, indicating its insufficient decomposition capability when handling arc signals.

Overall, these results validate the rationality of adopting DWT in this work.

2) *Cooperation of LFF and TFFI Modules:* As mentioned in Section II-B, LFF is to first align the multi-band feature signals through LrFC layers and then concatenate them to form the fused representation. Besides LrFC, possible alignment ways include the interpolation, Inverse DWT (IDWT), and FC, which all can be used in the LFF module. For feature interaction, we can replace the TFFI module with the DDI module or none.

Experimental results are shown in Table V, where “-” means no corresponding module. Under the same interaction condition, ADINet with LrFC always obtains the best performance, which shows LrFC has a good ability to preserve multi-band features in the fused representation. Under the same fusion way, ADINet with the TFFI module is always the best, which suggests that TFFI has a stronger time-frequency feature interaction capability than DDI does. Note that the performance of ADINet is further enhanced when LrFC and TFFI are joint, demonstrating the effectiveness and rationality of this combination for arc fault detection.

3) *Internal Structure of ADI Block:* To further evaluate the rationality of structural design in the ADI block of the TFFI module, we conducted ablation experiments. As stated in Section II-C, ADI deals with time and frequency information interaction, which is efficient and better than DDI according to Table V. Different from DDI, ADI has two special units: BAU and BGU, which are used to calculate band adaptive and gated weights, respectively. For band weights, we can give them in advance or by applying BAU. Three predefined configurations were $\mathbf{1} = [1, \dots, 1]^T \in \mathbb{R}^C$, $\mathbf{0.5} = 0.5[1, \dots, 1]^T \in \mathbb{R}^C$, and $\mathbf{0} = [0, \dots, 0]^T \in \mathbb{R}^C$, where $\mathbf{1}$ represents that the strength of all cross-band interaction information is maximum, and $\mathbf{0}$ means that the cross-band interaction information is ignored. For gated weights, two configurations were considered: the predefined vector $\mathbf{1}$ and an adaptive vector calculated by BGU, where $\mathbf{1}$ actually means no BGU.

Experimental results are provided in Table VI. Although the model obtains a rather high model performance (say 96.18% F1-score) with the predefined band weight vector $\mathbf{w}_b = \mathbf{1}$, it has a notable performance improvement (say 97.36% F1-score) when using BAU to adaptively calculate $\mathbf{w}_b$. After introducing BGU, the model shows its performance improvements across all configurations here. Most notably, when both BAU and BGU are enabled simultaneously, the model's performance experiences a significant leap, achieving the highest F1-score of 98.10% and accuracy of 99.15% among all configurations.

Above results demonstrate that BAU can dynamically adjust

TABLE III: Comparison of model performance on the IAED dataset.

Dataset	Method	F1-score (%)	Acc (%)	Dataset	Method	F1-score (%)	Acc (%)
HR-mix1	SVM	89.88 $\pm$ 0.53	95.67 $\pm$ 0.20	LR-mix2	SVM	86.84 $\pm$ 1.43	92.24 $\pm$ 0.85
	KNN	88.09 $\pm$ 0.98	95.00 $\pm$ 0.37		KNN	87.47 $\pm$ 0.63	92.62 $\pm$ 0.48
	RF	86.06 $\pm$ 3.93	93.64 $\pm$ 2.04		RF	89.12 $\pm$ 0.84	93.40 $\pm$ 0.60
	ArcNet	90.33 $\pm$ 0.68	96.00 $\pm$ 0.20		ArcNet	89.67 $\pm$ 0.91	93.91 $\pm$ 0.38
	GRU	94.14 $\pm$ 0.48	97.42 $\pm$ 0.22		GRU	89.80 $\pm$ 0.70	93.98 $\pm$ 0.28
	TNN	95.03 $\pm$ 0.28	97.89 $\pm$ 0.14		TNN	91.02 $\pm$ 1.01	94.58 $\pm$ 0.63
	MS-CNN	94.71 $\pm$ 0.54	97.76 $\pm$ 0.16		MS-CNN	91.57 $\pm$ 0.84	94.67 $\pm$ 0.44
	ADINet	97.32 $\pm$ 0.12	98.85 $\pm$ 0.06		ADINet	91.92 $\pm$ 0.40	95.15 $\pm$ 0.37
HR-single	SVM	91.50 $\pm$ 0.97	95.31 $\pm$ 0.47	LR-single	SVM	85.24 $\pm$ 0.76	89.88 $\pm$ 0.60
	KNN	91.70 $\pm$ 0.94	95.32 $\pm$ 0.52		KNN	86.59 $\pm$ 0.29	90.58 $\pm$ 0.17
	RF	92.24 $\pm$ 0.76	95.68 $\pm$ 0.40		RF	86.27 $\pm$ 0.29	90.30 $\pm$ 0.24
	ArcNet	93.31 $\pm$ 0.62	96.27 $\pm$ 0.29		ArcNet	86.13 $\pm$ 1.06	90.32 $\pm$ 0.71
	GRU	88.74 $\pm$ 0.47	95.12 $\pm$ 0.22		GRU	86.85 $\pm$ 0.83	91.11 $\pm$ 0.48
	TNN	96.23 $\pm$ 0.44	97.81 $\pm$ 0.28		TNN	87.42 $\pm$ 0.72	91.35 $\pm$ 0.50
	MS-CNN	92.71 $\pm$ 0.23	96.00 $\pm$ 0.20		MS-CNN	87.83 $\pm$ 0.79	91.64 $\pm$ 0.49
	ADINet	96.95 $\pm$ 0.34	98.23 $\pm$ 0.18		ADINet	94.77 $\pm$ 0.28	96.12 $\pm$ 0.26
HR-All	SVM	89.90 $\pm$ 0.49	95.26 $\pm$ 0.28	LR-all	SVM	84.87 $\pm$ 0.68	90.10 $\pm$ 0.47
	KNN	89.66 $\pm$ 0.32	95.13 $\pm$ 0.14		KNN	85.93 $\pm$ 0.77	90.56 $\pm$ 0.46
	RF	89.00 $\pm$ 0.57	94.78 $\pm$ 0.28		RF	85.79 $\pm$ 0.59	90.43 $\pm$ 0.34
	ArcNet	91.43 $\pm$ 0.46	96.02 $\pm$ 0.25		ArcNet	87.48 $\pm$ 0.41	91.69 $\pm$ 0.32
	GRU	94.38 $\pm$ 0.36	97.30 $\pm$ 0.15		GRU	88.61 $\pm$ 0.50	92.32 $\pm$ 0.40
	TNN	95.64 $\pm$ 0.43	97.91 $\pm$ 0.24		TNN	89.47 $\pm$ 0.35	92.64 $\pm$ 0.21
	MS-CNN	95.18 $\pm$ 0.21	97.71 $\pm$ 0.14		MS-CNN	88.31 $\pm$ 0.73	92.00 $\pm$ 0.48
	ADINet	97.90 $\pm$ 0.21	98.97 $\pm$ 0.11		ADINet	93.25 $\pm$ 0.39	95.20 $\pm$ 0.21

TABLE IV: Comparison of different signal transformation methods on the HR-mix1 subset.

Method	F1-score (%)	Accuracy (%)
FFT	96.90 $\pm$ 0.53	98.65 $\pm$ 0.26
PSD	92.89 $\pm$ 0.40	96.94 $\pm$ 0.23
STFT	96.80 $\pm$ 0.24	98.62 $\pm$ 0.09
CWT	92.96 $\pm$ 0.52	97.05 $\pm$ 0.20
HHT	81.40 $\pm$ 1.25	92.49 $\pm$ 0.54
WPT	87.78 $\pm$ 1.03	95.07 $\pm$ 0.44
DWT	97.32 $\pm$ 0.12	98.85 $\pm$ 0.06

TABLE V: Ablation results of feature alignment and interaction on the HR-mix1 subset.

Alignment	Interaction	F1-score (%)	Accuracy (%)
Interpolation	-	81.32 $\pm$ 0.59	92.53 $\pm$ 0.31
IDWT	-	87.61 $\pm$ 0.83	94.87 $\pm$ 0.37
FC	-	91.45 $\pm$ 0.11	96.33 $\pm$ 0.12
LrFC	-	97.32 $\pm$ 0.34	98.83 $\pm$ 0.14
Interpolation	DDI	90.53 $\pm$ 0.43	96.08 $\pm$ 0.21
IDWT	DDI	96.79 $\pm$ 0.40	98.58 $\pm$ 0.20
FC	DDI	95.44 $\pm$ 0.61	97.98 $\pm$ 0.23
LrFC	DDI	97.14 $\pm$ 0.44	98.72 $\pm$ 0.21
Interpolation	TFFI	90.99 $\pm$ 0.60	96.27 $\pm$ 0.19
IDWT	TFFI	97.09 $\pm$ 0.30	98.72 $\pm$ 0.10
FC	TFFI	95.51 $\pm$ 0.38	98.03 $\pm$ 0.18
LrFC	TFFI	98.10 $\pm$ 0.14	99.15 $\pm$ 0.06

the cross-band interaction strength according to the input signals, and BGU can effectively regulate the proportion of cross-band interaction information injected into the interacted time features, suppressing redundant residual signals.

TABLE VI: Ablation Study Results of BAU and BGU on the HR-mix1 Subset

Band weights	Gated weights	F1-score (%)	Accuracy (%)
1	1	96.18 $\pm$ 0.28	98.37 $\pm$ 0.16
0.5	1	93.84 $\pm$ 0.41	97.41 $\pm$ 0.17
0	1	94.62 $\pm$ 0.33	97.63 $\pm$ 0.12
BAU	1	97.36 $\pm$ 0.32	98.85 $\pm$ 0.14
1	BGU	96.42 $\pm$ 0.32	98.44 $\pm$ 0.17
0.5	BGU	96.11 $\pm$ 0.36	98.34 $\pm$ 0.17
0	BGU	95.95 $\pm$ 0.31	98.22 $\pm$ 0.11
BAU	BGU	98.10 $\pm$ 0.14	99.15 $\pm$ 0.06

E. Model Complexity Analysis

To evaluate the model complexity of the proposed ADINet, we made a comparative analysis on inference time and parameter count against representative deep models, including ArcNet, GRU, TNN, and MS-CNN. All experiments here were performed on the HR-mix1 subset.

Comparison of five deep models is illustrated in Fig. 6, where the horizontal coordinate represents the inference time (s) of models, the vertical coordinate reflects the accuracy of models, and the size of circles filled with various colors denotes the parameter count (M). From Fig. 6, we can see that ADINet achieves the highest accuracy and the smallest

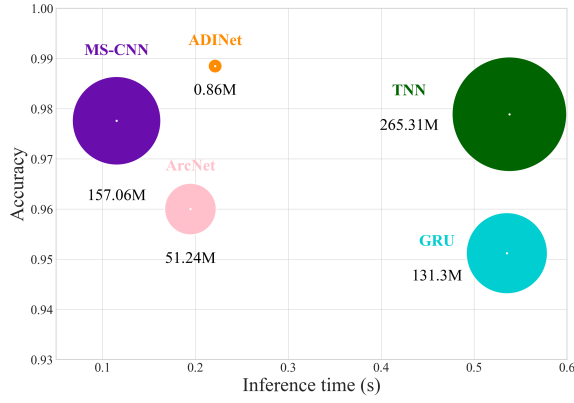


Fig. 6: Model complexity analysis of five deep models.

parameter count among all compared models, and maintains a competitive inference speed. Although the inference speed of ADINet is slightly slower than that of both MS-CNN and ArcNet, ADINet contains only 0.86M parameters, approximately 60 and 180 times fewer than ArcNet (51.24M) and MS-CNN (157.06M), respectively. Relatively, ADINet is more suitable for deployment on embedded platforms with limited computing resources than both MS-CNN and ArcNet. Both GRU and TNN models have not only large parameter counts and slow inference speed but also achieve a lower accuracy than ADINet. These results suggest that heavy sequence-based or attention-based architectures do not necessarily provide advantages for arcing fault detection.

Simply speaking, the low model complexity of ADINet is mainly due to the structural design of LrFC and TFFI modules. For example, the computational complexity of an LrFC block is $O(\ell/2(\ell + L/2))$ when the input vector of this block has the length of ℓ . If we replace the LrFC block with one FC block, then the computational complexity is $O(\ell L/2)$. Because we assume that $\ell < L/2$, the computational complexity of the LrFC block is much lower than that of the FC block. The extremely compact parameter footprint of ADINet enables efficient memory usage and low computational overhead, while the adaptive time-frequency interaction through the TFFI module ensures robustness across diverse load conditions. In a nutshell, ADINet has a superior performance-efficiency profile that aligns well with the real-time and resource-constrained requirements of practical arc fault protection devices.

IV. CONCLUSION

This study proposes ADINet, an arc fault detection network that integrates LFF and TFFI modules. The LFF module employs LrFC layers to efficiently fuse multi-band features while preserving both low-frequency steady-state information and high-frequency transient characteristics. The TFFI module enables time and frequency-band information interaction through ADI blocks equipped with BAU and BGU.

Comprehensive experiments on the IAED dataset demonstrate that ADINet consistently achieves the best detection performance across all subsets, with F1-scores of 96.95%–97.90% on the HR set and 91.92%–94.77% on the LR set. Ablation experiments demonstrate that LrFC outperforms conventional

alignment ways and the efficiency of TFFI module. For the structure design of ADI, we also show the indispensability of two new units (BAU and BGU) through ablation. Moreover, ADINet contains only 0.86M parameters, approximately 60 and 180 times fewer than ArcNet and MS-CNN respectively, making it highly suitable for resource-constrained embedded platforms.

Future work will focus on enhancing the model's adaptability to more complex load types and exploring its deployment in industrial arc fault detection scenarios.

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